



**National College of Ireland
Project Submission Sheet**

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Lecturer: Prof. Eric Gyamfi

Submission

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I hereby certify that the information contained in this (my submission) is information pertaining to research I conducted for this project. All information other than my own contribution will be fully referenced and listed in the relevant bibliography section at the rear of the project.

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Signature: Mohd Nizam

Date: 28/12/2025

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2. Projects should be submitted to your Programme Coordinator.
3. **You must ensure that you retain a HARD COPY of ALL projects**, both for your own reference and in case a project is lost or mislaid. It is not sufficient to keep a copy on computer. Please do not bind projects or place in covers unless specifically requested.
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AI Acknowledgement Supplement:

Module: Programming for Financial Data Science (MSCFTD1)

Your Name/Student Number	Course	Date
Mohd Nizam Shaikh / 24198170	Master of Science in FinTech	28/12/2025

This section is a supplement to the main assignment, to be used if AI was used in any capacity in the creation of your assignment; if you have queries about how to do this, please contact your lecturer. For an example of how to fill these sections out, please click [here](#).

AI Acknowledgment:

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool
N/A	N/A	N/A

Description of AI Usage:

This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how these responses were utilized or modified in the assignment. **One table should be used for each tool used.**

Evidence of AI Usage:

This section includes evidence of significant prompts and responses used or generated through the AI tool. It should provide a clear understanding of the extent to which the AI tool was used in the assignment. Evidence may be attached via screenshots or text.

Additional Evidence:

[N/A]

Academic Honesty Declaration

I declare the following to be true for this submission:

- I have completed the task during the designated time window and declare it to be exclusively my own work.
- I have not received, or attempted to receive assistance in preparing this response from any other person during the assessment window.
- I have not provided, or offered to provide, assistance to any other student by any means during the assessment window.
- I have read and understand the National College of Ireland guidelines of Plagiarism.

PROGRAMMING FOR FINANCIAL DATA SCIENCE

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PROJECT TITLE: The **STRATOS** [*Strategical Tactical Response and Targeting Operating System*]

I. **PRESENTATION URL** – https://studentncirl-my.sharepoint.com/:v:/r/personal/x24198170_student_ncirl_ie/Documents/Recordings/Meeting%20with%20Mohd%20Nizam%20Mohd%20Nasir%20Shaikh-20251228_163102-Meeting%20Recording.mp4?csf=1&web=1&e=oezIB4&nav=eyJyZWZlcnJhbEluZm8iOmsicmVmZXJyYWxBcHAIoiJTdHJIYW1XZWJBcHAIcCJyZWZlcnJhbFZpZXciOiJTaGFyZURpYWxvZy1MaW5rIiwicmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYiIsInJlZmVycmFsTW9kZSI6InZpZXcifX0%3D

II. POWERPOINT PPT URL -

https://docs.google.com/presentation/d/1xmGdvSJ406iWde59enY3yaT4Yp1Hpdqg/edit?usp=drive_link&oid=111490828299032735772&rtpof=true&sd=true

III. PROJECT REPORT

PART A: PROBLEM DEFINITION AND DATA ACQUISITION

A.1. BUSINESS CASE:

Executive summary: The project “**STRATOS**” is an advanced Data Analytics and Machine Learning pipeline simply created for IRISH – i police force to assign officers and resource allocation when crime rates are fluctuating plus where actually necessary. I analysed crime offence trends in various regions and quarters based from my dataset which is CJQ05 provided by Central Statistics Office (CSO) to overview patterns. After that, I developed and tested many different codes to train and tested machine learning models to forecast future crime rates with that I got 99% accuracy score. I discovered that random forest model was the best model to predict further crime waves which gave law enforcement a useful tool to plan shifts and resources in high crime sector before any further delays responding to incidents.



Core challenge - Ireland Crime rates are continuously changing based on the time of year, region and multiple social issues. Predicting the exact moment and place of criminal activities growth is a significant difficulty for the Garda because of that it is challenging to match actual police task force demand and supply in absence of accurate forecasting technologies. Though, this results in inefficiencies such as understaffed in required conflict regions and overstaffed in quiet regions. Such discrepancy can lead to risk by compromising public safety via slowed down responding time which can be a big crucial asking of wasting public funds and valuable resources.

Aim – Simply by creating a machine learning with effective models it can provide a better solution that will forecast future crime rates which has an objective to fix the resource imbalance. We have capability to predict accurate forecasting for crimes across for all four quarters and locations by deep studying and investigating of patterns from past data. Law enforcement authority can be able to look into the product and shifts from reactive to proactive actions. At the end, this can help to update the roaster and makes it possible to strategically

schedule shifts and deploy resources which remarks that police are patrolling right when they needed to prevent crime and gave security to public.

A.2. ML TASK & TARGET:

ML Task for this specific project is “***Regression***” because of the nature of our objective we are not attempting to categorize localities as safe or risky which can be easily done by classification. Rather we going to forecast realistic numbers of crime incident which are likely to occur additionally predicted by regression models using features like location and time. The target variable (Y) is VALUE column for our analysis as it is fundamental truth in our dataset. It showcases the overall number of registered criminal activities for a kind of offense happened in the particular Garda Region over the quarter of each year. In order to figure out future date values with these numbers when the actual crime data is still unknown, basically the algorithm will learn relation between other variables during modelling phase.

A.3. DATA ACQUISITION:

The dataset source for this project is collected by Irish government’s open data portal from the CSO database which has name given onsite as CJQ05 - Recorded Crime Incidents and link is <https://data.gov.ie/dataset/cjq05-recorded-crime-incidents>. As this dataset provides crime record statistics in Ireland covering several years with the structured view of criminal activities organized geographically.

The major columns of the dataset to predict future Values are as follows:

- Quarter – Allow us to monitor seasonal trends by highlighting the 3 months period of each year from Q1 of 2003 till Q2 of 2025.
- Garda Region – It defines the geographic area of 4 regions which are Dublin Metropolitan, North Western, Eastern & Southern Region which helps to analysis crime in different region.
- Type of Offence – This provides a structural format of many crime types occurred or not with their sections.
- VALUE – The numerical data that describes how many times offences were actually reported in each Garda Region.

The main highlighted key features including the patterns and trends which required to train models efficiently, as giving access to this historical data is significant and it would be difficult to make exact forecasts about the future without the depth of this historical data.

PART B: DATA ANALYSIS AND PRE-PROCESSING

B.1. EXPLORATORY DATA ANALYSIS:

	TLIST(Q1)	C02479V04377	VALUE
count	25920.000000	25920.000000	25872.000000
mean	20140.033333	25.000000	412.008619
std	64.956357	11.180556	900.306919
min	20031.000000	10.000000	0.000000
25%	20083.000000	17.500000	4.000000
50%	20141.500000	25.000000	31.000000
75%	20194.000000	32.500000	401.000000
max	20252.000000	40.000000	10475.000000

I began analysis by looking at the types and structure of the data to learn and better understand the dataset before training, testing models and metrics. I observed that the dataset contains more than 25000 rows and 10 columns at the beginning. It has a combine value of numerical and text data, for instance columns such as Quarter, Garda region & Type of offence.

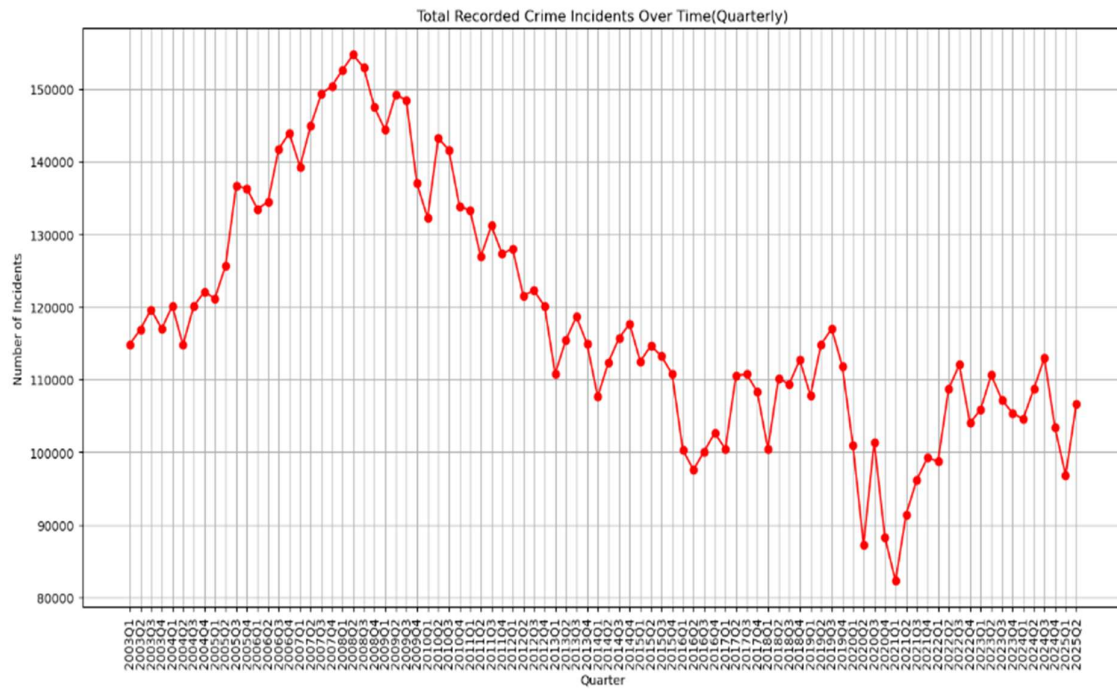
(25920, 10)

	STATISTIC	Statistic Label	TLIST(Q1)	Quarter	C02479V04377	Garda Region	C02480V03003	Type of Offence	UNIT	VALUE
0	CJQ05	Recorded crime incidents	20031	2003Q1	10	10 Dublin Metropolitan Region	1	Homicide & related offences (01)	Number	7.0
1	CJQ05	Recorded crime incidents	20031	2003Q1	10	10 Dublin Metropolitan Region	111	Murder (0111)	Number	4.0
2	CJQ05	Recorded crime incidents	20031	2003Q1	10	10 Dublin Metropolitan Region	112	Manslaughter (0112)	Number	2.0
3	CJQ05	Recorded crime incidents	20031	2003Q1	10	10 Dublin Metropolitan Region	113	Infanticide (0113)	Number	0.0
4	CJQ05	Recorded crime incidents	20031	2003Q1	10	10 Dublin Metropolitan Region	12	Dangerous driving leading to death (012)	Number	1.0

I used summary statistics because when in investigating period I noticed that there are odd numbers in my crime VALUE target variable with a broad range of null values, as there are thousands of incidents in some of the regions but nearly no instance found out in other regions. In order to investigate how crimes are actually fluctuating it was confirmed that most of the recorded crime cases were on the lower end with a few high spikes over the year and quarter, in addition in various Garda Regions, so I utilized four types of plots which clearly indicated trends are as follows:

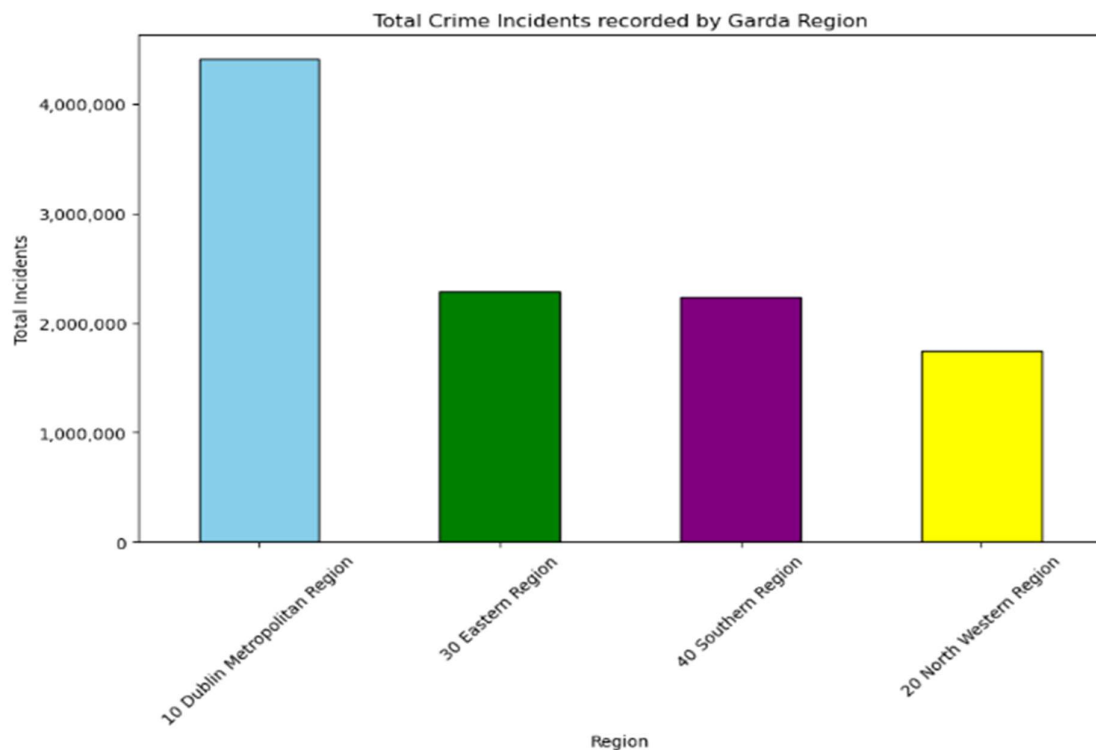
i. *Fig.1. Time Series Plot by Year & Quarter:*

The below line chart shows the total number of crimes over the number of years with that visualization we can immediately know whether the overall trend is increasing or decreasing. It also clearly highlights the ups and downs between each quarter which proves that crime is a major seasonal trend rather than a random incident.



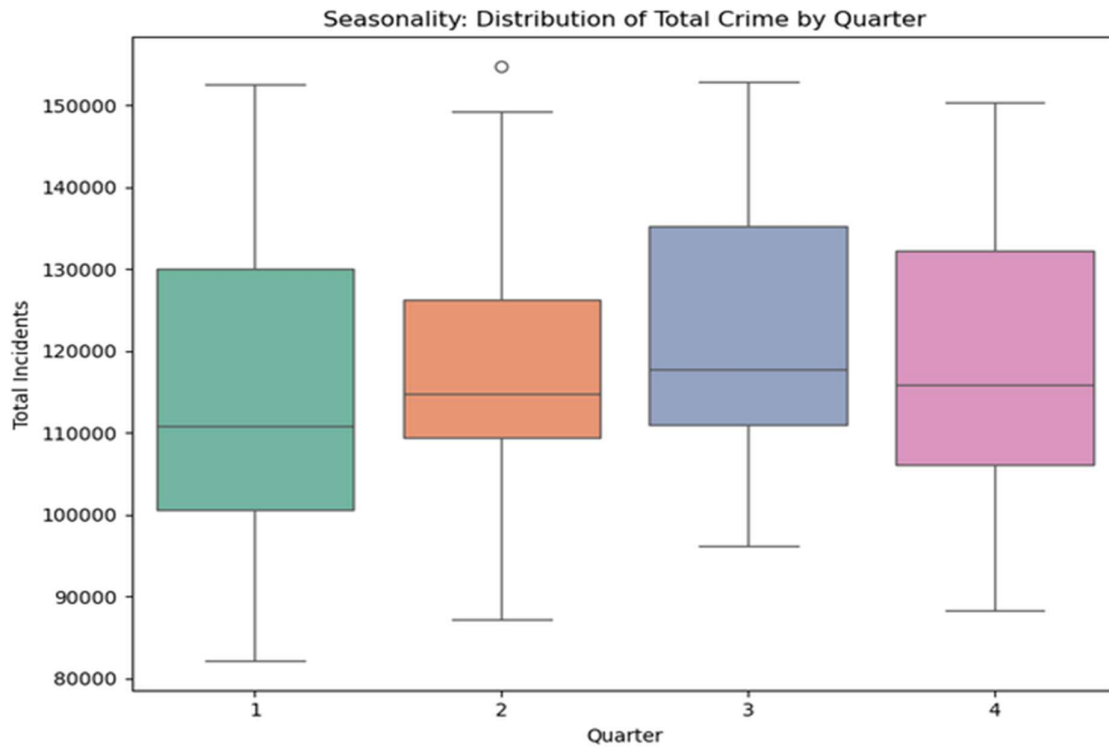
ii. Fig.2. Bar Graph for Total Incidents by Regions:

This bar chart will allow us to compare the overall number of crimes committed in each of the four Garda Regions and it is easy to view that Dublin Metropolitan Region has the highest number of crimes. It describes the imbalance in the dataset proving that metropolitan areas have substantial workload.



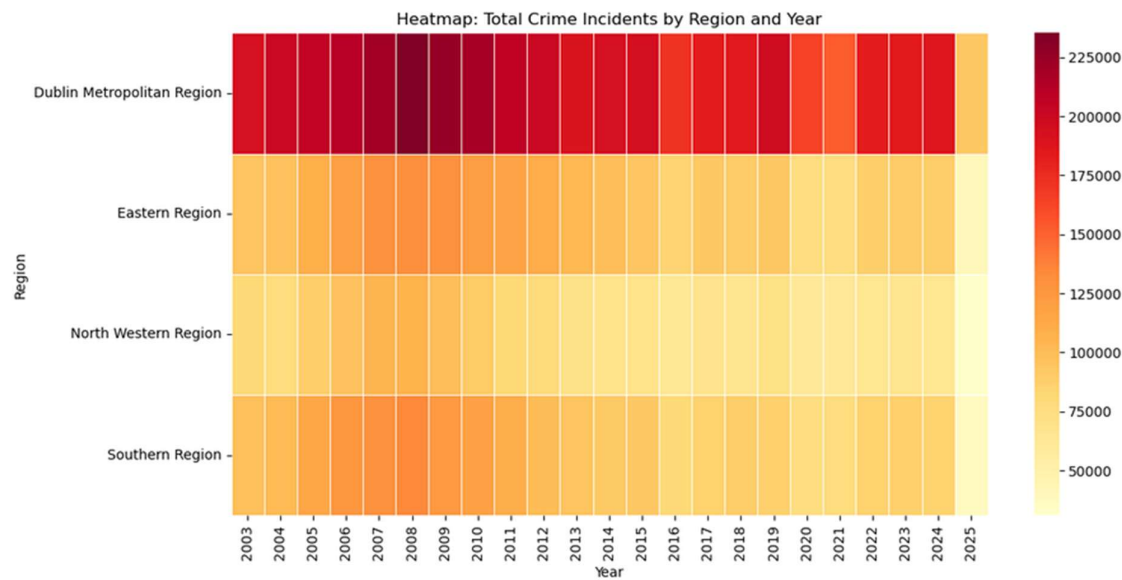
iii. *Fig.3.Box Plot by Seasonal Analysis:*

The Box plot used to present visualization for the level of crime in each season and identify spikes in data by breaking down the data in a set of quarters. It will help us to understand whether crime is higher or lower which is during specific times of year such as in winter or in summer months.



iv. *Fig.4.Heatmap chart of Region by Year:*

Overall, this heatmap graphic uses colour intensity to track the risky area allowing us to see in a single view as which regions were the most dangerous during the specific period of year. It provides a quick way to compare several places at once and immediately illustrates if a region is safer which is light colour or worse which is in darker colour.



B.2. MISSING DATA:

The most significant VALUE variable for prediction had some of the missing values I got to know in the period of auditing data, since removing rows will alternatively result in loss of important data because this data contains crime counts give errors in model prediction accuracy. So, I decided to use column median values to fill the gaps of null in between of rows because crime data contains large spikes of significant outliers that would alter the average due to that I choose median rather than average because it represents realistic middle ground.

<code>df.isnull().sum()</code>		<code>df["VALUE"] = df["VALUE"].fillna(df["VALUE"].median())</code>	
STATISTIC	0	<code>df.isnull().sum()</code>	
Statistic Label	0	STATISTIC	0
TLIST(Q1)	0	Statistic Label	0
Quarter	0	TLIST(Q1)	0
C02479V04377	0	Quarter	0
Garda Region	0	C02479V04377	0
C02480V03003	0	Garda Region	0
Type of Offence	0	C02480V03003	0
UNIT	0	Type of Offence	0
VALUE	48	UNIT	0
dtype: int64		VALUE	0
		dtype: int64	

Additionally, I dropped some columns like “STATISTIC”, “Statistic Label”, “TLIST(Q1)”, “C02479V04377”, “C02480V03003” & “UNIT” as it contains identical category wordings and duplicate values which is available in another useful columns because it doesn’t offer machine learning model valuable information.

```
# Dropping below columns which not required for ML model
df = df.drop(columns=["STATISTIC", "Statistic Label", "TLIST(Q1)", "C02479V04377", "C02480V03003", "UNIT"], axis=1)
```

```
df.head()
```

	Quarter	Garda Region	Type of Offence	VALUE
0	2003Q1	10 Dublin Metropolitan Region	Homicide & related offences (01)	7.0
1	2003Q1	10 Dublin Metropolitan Region	Murder (0111)	4.0
2	2003Q1	10 Dublin Metropolitan Region	Manslaughter (0112)	2.0
3	2003Q1	10 Dublin Metropolitan Region	Infanticide (0113)	0.0
4	2003Q1	10 Dublin Metropolitan Region	Dangerous driving leading to death (012)	1.0

Furthermore, I have removed columns which were firstly utilized for some EDA than after splitting the features for further objective to train models effectively. This is the pure analysis which will help to train models effectively although in the future, it will provide precise prediction to provide absolute outcomes.

```
df = df.drop(columns=["Quarter", "Garda Region", "Type of Offence", "Region Code", "Offence Section"], axis=1)
```

```
df.tail()
```

	VALUE	Year	Qtr Num	Region	Offence
25915	578.0	2025	2	Southern Region	Offences against government, justice procedure...
25916	18.0	2025	2	Southern Region	Offences against government and its agents
25917	0.0	2025	2	Southern Region	Organisation of crime and conspiracy to commit...
25918	3.0	2025	2	Southern Region	Perverting the course of justice
25919	557.0	2025	2	Southern Region	Offences while in custody, breach of court ord...

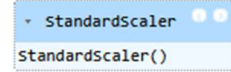
B.3. FEATURE ENGINEERING:

As Machine Learning model algorithms can't identify categorical data, so I had changed that into numerical data thus depending on the types of features I applied two different methods. I applied One-Hot Encoding for Region using get_dummies code for small variables where there is no prior set order to ensure that the model considers each place of region equally. So, it doesn't assume a mathematical ranking value between them as this method help to generate binary columns such as 0 or 1 for each region because there is small number of categorical values. On the other hand, I converted column of Offence which has a lot of various categories or large number of unique categories to transform into simple numbers for example 0, 1, 2, etc.

	VALUE	Year	Qtr Num	Offence	Region_Dublin Metropolitan Region	Region_Eastern Region	Region_North Western Region	Region_Southern Region
24963	6.0	2024	3	24	0	1	0	0
8989	3518.0	2010	4	18	1	0	0	0
15730	1075.0	2016	3	8	0	1	0	0
9515	117.0	2011	2	29	1	0	0	0
9906	1438.0	2011	3	48	0	0	1	0

B.4. STANDARDIZATION SCALING:

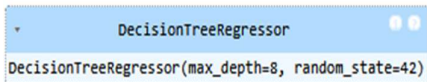
Lastly, I utilized feature scaling with the Standard Scaler method to make sure every feature of my final dataset posture contributes to the model because I need to make sure huge value of year column doesn't bypass encoded Region feature which has a few numbers. I split all of the encoded and numerical features into (X) and VALUE Label into (y) for 80% training and 20% testing with the help of train, test & split to transform X & y into scaler fit. As this give clarification to data distribution and via accurate inputs regression algorithms works stable through the learning process.



PART C: MODEL IMPLEMENTATION AND TRAINING

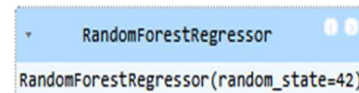
C.1. TRADITIONAL MODEL: DECISION TREE REGRESSOR:

Traditional models are the key implementation which will give a significant booster start to ML and I have chosen the Decision Tree Regressor which is the best model for analysis and training as per the characteristics of crime offence data. As it is good at capturing non-linear relationship between features that linear model would miss some of the them such as the different variety of crime rates would increase and decrease in various quarters and regions. Basically, I started model through limiting the tree depth by *max_depth=8* as I did it intentionally to encourage tree to learn from general trends and not to memorize every single bit of noise to prevent overfitting in the training set. So, to assure my results are logically reproducible each time the code runs, I also entered *random_state=42*. The Decision tree had a higher error rate compared to advanced models as it is illustrating limits in forecasting accuracy of crime numbers on unseen future data however as a result it was able to capture the key trends.



C.2. ADVANCED MODEL: RANDOM FOREST REGRESSOR:

For improved prediction accuracy I selected Random Forest Regressor as my advanced model because it works as the backbone of ML to train and give better outcomes. Compared to decision tree this technique is great for enhancing the prevention of overfitting an increases stability by building multiple decision trees during the training and combining the output for better results. I putted *random_estimator=100* for model to build 100 separate trees to provide the finalized forecast additionally for better consistency, I kept the *random_state=42*. Hence, the advanced strategy is better for the crime offense dataset because it manages all the complicated relationships between offense types, time and location.



C.3. CODE QUALITY:

To ensure the Jupyter notebook looks clean and good I followed high code quality through the project guidelines to make sure it is simple but effective to operate. As I logically arrange the dataset firstly through data loading, then followed by imputing median values & cleaning some features, after that completed pre-processing for visualization and model training plus testing for lastly to evaluation of metrics for prediction accuracy. The code structure was self-explanatory as I used simple variable names for each code as it can be easily clarified each of them for example *Crime_time_series* and *rf_model*, although additionally I also provided adequate comments that explains the logic behind every certain actions. If in future any other data scientist or analyst may go through the notebook and will understand the fully process of

analysis without further explanation because I think of its format and explanation of each and every steps.

PART D: EVALUATION, COMPARISON & REPORT

D.1. METRIC SELECTION: RMSE (ROOT MEAN SQUARED ERROR) AND R-SQUARED (R2):

Metrics selection are the most crucial part for ML models as it measures quantitative scores of how well the models are performing for each task for as predicting accurately. In this analysis, I have carefully selected RMSE (Root Mean Squared Error) and R2 (R-Squared) because it is the main evaluation metrics for my crime offence dataset as RMSE is especially effective for crime data with sudden surge in crime. Since it is risky to underestimate a majority of crimes than a few errors in forecasting a calm day so RMSE highlights large errors more strongly than smaller ones as this is essential for Gardai. It gives us the error of same unit as data for example number of events which is easily explainable to stakeholders. R2 provides percent of variation showing model fit & explanatory power because crime is a complex process that involves variable like seasons & locations as a result high score means models aren't just guessing but in fact it is understanding the roots of dataset.

RMSE for Decision Tree Regressor: 315.32 RMSE for Random Forest: 78.72

R2 Score for Decision Tree Regressor: 0.8830 R2 Score for Random Forest: 0.9927

D.2. COMPARATIVE RESULTS:

In order to present a fair comparison between both the models I have used same dataset in that notebook to train both models and after the deployment of metrics in unobserved test set to compare them. The results are demonstrated in clear manner that performance differ between both the evaluation metrics as random forest outsmart decision tree model based on the accurate prediction as describe details below in table format:

Models	RMSE	R-Squared
Decision Tree Regressor	315.32	0.8830
Random Forest Regressor	78.72	0.9927

To summarize about the model decision tree had a little higher RMSE-315.32 and on the other hand lower score of R2 was 88% as compared to another model. Although Random Forest was far more effective than Decision Tree but it was respectable at initial stage however, with an R2 of nearly 100% Random Forest were able to explain almost all of the variation shifts in data. Additionally, it is my sole option for future which can be useful when I deployed the project of “STRATOS” in real world due to its forecast which is significant in line with actual crime numbers due to smaller RMSE-78.72 furthermore it will be better implication to government authorities for safeguarding public with endurable security.

D.3. MODEL DISCUSSION:

The better performance of Random Forest model as compared to the Decision Tree model can be explained by the structure plus design of algorithms and type of data in dataset. In order to understand the characteristics and variation of training data, any specific Decision Tree has a substantial probability of overfitting which leads to create complex rules. When it will look

into new data or test sets it will probably struggle with generalization which results to higher RMSE but in the other hand Random Forest solves problem by utilizing techniques called as “*Bagging*” which basically involves building 100 trees based on random selection data samples and after that does prediction by averaging. As a result, the dataset of Crime Offense includes multiple offence types and various regions thereby compounding the problem of high-dimensional data resulting the Random Forest’s ability to handle such high-dimensional data without overfitting the model is important to achieving a higher accuracy score than Decision tree. This technique makes the group of trees smoother as well as fewer errors.

D.4. BUSINESS IMPACT:

Deploying the Random Forest model would definitely change the nature of operations of *An Garda Siochana* through becoming more proactive rather than reactive as in current situation resources are only occasionally allocated by guessing and previous incidents which doesn’t work in its full potential and might result in inefficiencies. The head operations could have accurately predicted the level of demand for the upcoming quarters if it will have a model in ML that predicts score of crimes with 99% above accuracy. Whenever resources of patrol cars and policemen are patrolling in certain stationed areas at the exact time when it’s predicted the crime would be at highest, this helps to deliver for *Strategic Resource Allocation* which over “*STRATOS*” project will does in future.

Return on Investment (ROI): The ROI would be achieved once after the implementation via cost reduction and increased safety & security. The force can reduce unnecessary instances such as overtime spending in areas with low crime rates which will be effective by scheduling shifts. Moreover, the operational change has an impact of having police force availability when crimes occur which is specifically identified an obstacle before but when project strategy where fully implemented it may eventually lower overall crime rates. To sum up, it will improve public safety in the neighbourhood and trust among the governmental authority as well as it reaches to the objective of organization.

CONCLUSION:

To conclude this project has proven that machine learning has ability to forecast crime in Ireland efficiently using previous historical information supplied by the CSO, I developed a Random Forest model that can literally forecast the crimes incident of over 99% accurately. It showcases that crime does not happen randomly but rather indicates pattern based on location and season. The model provides a proactive policing tool as the project greatest achievement will be that it proves how informational data can be used to enhance effective allocation of police resources, placing officers at the appropriate frequency instead of waiting for emergency calls as law enforcement can be able to utilize these predictions for placing officer at actual place before any incidents takes place which saves money and improves safety.

FUTURE WORK:

While my project models are in high level accuracy, to deploy this tool in real world on the hands of police force I would specifically enhance three major important areas which will be following below:

- *Time-split Validation*: At present model it was tested using a random split of data but for real world it would be different as we cannot shuffle time however, I trained on data by

year 2003 to 2023 and tested by year 2024 to 2025. As this is to test model's ability to predict the unknown future more effectively.

- *Analysing of external factor:* Crime does not exist in an empty space for such further work need to involve merging of this dataset with other features like on any weather conditions could reduce crime or unemployment rates and major events for example concerts or matches.
- *Deployment:* The project existing models are forecasting crime by regions but to make this useful for local officer I would filter likely to filter data by each Garda Station so that it will be easily accessible for them to utilize. Finally, I will include this code into an easy-to-use dashboard so that cops would only need to choose their location to look for a crime forecast graph on tab without having a knowledge of python programming.

****THANK YOU****